



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

APPLIED PHYSICS DEGREE PATHWAYS

Note: This document is intended to serve as a scheduling aid for Applied Physics students. The schedule information provided is based on historical patterns and expected scheduling. Course offerings and modalities are subject to change due to instructor availability, student demand, and program priority. Students should check SIS for official course schedule information and the Johns Hopkins University Academic Catalog for official degree requirements.

APPLIED PHYSICS CORE COURSES & ELECTIVES | GENERAL M.S. DEGREE

(10) Courses: (4) Core Courses [at least (4) from the first six listed] + (6) Electives [including at least (3) 700- or 800-level]

CORE COURSES

615.641
Mathematical Methods
for Physics & Engineering
Every Term • Asynchronous

615.653
Classical Mechanics
Every Term • Asynchronous

615.671
Principles of Optics
Every Term • Asynchronous

615.642
Electromagnetics
FA / SP • Asynchronous

615.654
Quantum Mechanics
FA / SP • Asynchronous

615.680
Materials Science
FA / SP • Asynchronous

615.651
Statistical Mechanics
and Thermodynamics
Every Term • Asynchronous

615.665
Modern Physics
SP / SU • Asynchronous

675.613
Bold Science
Every Term • Asynchronous

615.661
Introduction to
Planetary Science
SP / SU • Asynchronous

615.662
Introduction to
Astrophysics
FA • Asynchronous

615.663
Physics of
Lunar Exploration
In Development • Async.

615.664
Introduction to
Exoplanets
In Development • Async.

615.666
Earth Systems & AI
In Development • Async.

615.682
Materials Characterization
In Development • Async.

615.683
Materials in
Extreme Environments
In Development • Async.

615.685
Quantum Measurement
and Sensing
In Development • Async.

615.715
Applied Atomic
and Molecular Physics
Spring • Asynchronous

615.744
Physics of
Space Systems I
SU / FA • Asynchronous

615.745
Physics of
Space Systems II
SP • Asynchronous

615.747
Sensors and
Sensor Systems
FA / SP • Asynchronous

615.748
Introduction to Relativity
SP • Asynchronous

615.749
Applied Nuclear Physics
In Development • Async.

615.751
Modern Optics
Every Term • Asynchronous

615.753
Plasma Physics
SU / FA • Asynchronous

615.755
Space Physics
FA / SP • Asynchronous

615.757
Solid State Physics
FA • Asynchronous

615.760
Physics of
Semiconductor Devices
SP • Asynchronous

615.761
Introduction to
Oceanography
SP • Asynchronous

615.762
Applied Computational
Electromagnetics
SP • Synchronous Online

615.763
Infrared Sensors:
Physics and Applications
SP • Asynchronous

615.765
Chaos and Its
Applications
SU / FA • Asynchronous

615.769
Physics of
Remote Sensing
SP • Asynchronous

615.772
Cosmology
SU • Asynchronous

615.775
Physics of Climate
SP • Asynchronous

615.776
Quantum Algorithms:
Near Term Applications
SU / SP • Asynchronous

615.778
Optical System Design
and Modelling
Every Term • Asynchronous

615.780
Optical Detectors
and Applications
FA • Asynchronous

615.781
Quantum Information
Processing
SP • Asynchronous

716.782
Optics and MATLAB
SP • Asynchronous

615.783
Experimental
Quantum Computing
In Development • Async.

615.784
Superconducting Devices:
Physics and Applications
FA / SU • Asynchronous

615.786
Introduction to
Open Quantum Systems
In Development • Async.

615.800
Applied Physics Project
Every Term • Asynchronous

615.802
Directed Studies in
Applied Physics
Every Term • Asynchronous

MATERIALS & CONDENSED MATTER CONCENTRATION

(4) Applied Physics Core Courses + (6) MCM Electives, including at least (3) 700-level courses

CORE COURSES

615.641
Mathematical Methods
for Physics & Engineering
Every Term • Asynchronous

615.642
Electromagnetics
FA / SP • Asynchronous

615.651
Statistical Mechanics
and Thermodynamics
Every Term • Asynchronous

615.680
Materials Science
FA / SP • Asynchronous

Unshaded boxes indicate elective courses not offered by the Applied Physics program. Check the EP website / SIS for course offering information.

Electives

615.654
Quantum Mechanics
FA / SP • Asynchronous

615.682
Materials Characterization
In Development • Async.

615.683
Materials in
Extreme Environments
In Development • Async.

515.617
Nanomaterials

515.635
Mechanical Properties
of Materials

525.606
Electronic Materials

525.621
Introduction to
Electronics & the Solid
State

535.684
Modern Polymeric
Materials

700-level courses – Select at least 3

615.715
Applied Atomic
and Molecular Physics
Spring • Asynchronous

615.747
Sensors and Sensor
Systems
FA / SP • Asynchronous

615.757
Solid State Physics
FA • Asynchronous

615.760
Physics of
Semiconductor Devices
SP • Asynchronous

615.763
Infrared Sensors:
Physics and Applications
SP • Asynchronous

615.784
Superconducting Devices:
Physics and Applications
FA / SU • Asynchronous

615.800
Applied Physics Project
Every Term • Asynchronous

615.802
Directed Studies in
Applied Physics
Every Term • Asynchronous

535.732
Fatigue and Fracture
of Materials

535.748
Stress Waves, Impacts,
and Shockwaves

APPLIED OPTICS & PHOTONICS CONCENTRATION

(4) Applied Physics Core Courses + (1) ECE Core Course + (3) 700-level Electives + (2) Additional Electives

CORE COURSES

Applied Physics

Select (4) of the following

615.641
Mathematical Methods
for Physics & Engineering
Every Term • Asynchronous

615.642
Electromagnetics
FA / SP • Asynchronous

615.653
Classical Mechanics
Every Term • Asynchronous

615.654
Quantum Mechanics
FA / SP • Asynchronous

615.665
Modern Physics
SP / SU • Asynchronous

615.671
Principles of Optics
Every Term • Asynchronous

Unshaded boxes indicate elective courses not offered by the Applied Physics program. Check the EP website / SIS for course offering information.

CORE COURSES

Electrical & Computer
Engineering

Select (1) of the following

525.613
Fourier Techniques
in Optics

525.625
Laser Fundamentals

525.691
Fundamentals
of Photonics

* Students interested in taking 615.653/615.654 must complete 615.641 as a prerequisite

ELECTIVES

Select at least (3) of the following 700-level courses

615.751
Modern Optics
Every Term • Asynchronous

615.778
Optical System Design
and Modelling
Every Term • Asynchronous

615.780
Optical Detectors
and Applications
FA • Asynchronous

615.781
Quantum Information
Processing
SP • Asynchronous

716.782
Optics and MATLAB
SP • Asynchronous

525.753
Laser Systems
and Applications

525.756
Optical Propagation,
Sensing, and Backgrounds

525.772
Fiber Optic
Communication Systems

525.797
Advanced
Fiber Optic Laboratory

525.796
Introduction to
High-Speed Optoelectrics

585.734
Biophotonics

SPACE AND EARTH SCIENCE FOCUS AREA

(4) Space Science Core Courses + (2) Earth Science, (2) Heliophysics & Planetary Science, (2) Astrophysics Electives

CORE COURSES

Select 4 of the following,
including 675.613, followed by 615.744

Required

675.613
Bold Science
Every Term • Asynchronous

615.641
**Mathematical Methods
for Physics & Engineering**
Every Term • Asynchronous

675.744
**Physics of
Space Systems I**
SU / FA • Asynchronous

615.642
Electromagnetics
FA / SP • Asynchronous

615.653
Classical Mechanics
Every Term • Asynchronous

615.654
Quantum Mechanics
FA / SP • Asynchronous

615.665
Modern Physics
SP / SU • Asynchronous

615.671
Principles of Optics
Every Term • Asynchronous

615.651
**Statistical Mechanics
and Thermodynamics**
Every Term • Asynchronous

615.680
Materials Science
FA / SP • Asynchronous

EARTH SCIENCE ELECTIVES

Select 2 of the following

615.666
Earth Systems & AI
In Development • Async.

615.761
**Introduction to
Oceanography**
SP • Asynchronous

615.769
**Physics of Remote
Sensing**
SP • Asynchronous

615.775
Physics of Climate
SP • Asynchronous

HELIOPHYSICS AND PLANETARY SCIENCE ELECTIVES

Select 2 of the following

615.661
**Introduction to
Planetary Science**
SP / SU • Asynchronous

615.663
**Physics of
Lunar Exploration**
In Development • Async.

615.664
**Introduction to
Exoplanets**
In Development • Async.

615.745
**Physics of
Space Systems II**
SP • Asynchronous

615.755
Space Physics
FA / SP • Asynchronous

ASTROPHYSICS ELECTIVES

Select 2 of the following

615.662
**Introduction to
Astrophysics**
FA • Asynchronous

615.748
Introduction to Relativity
SP • Asynchronous

615.772
Cosmology
SU • Asynchronous

HARDWARE ELECTIVES

OPTIONAL: Select up to 2
of the following to
substitute other electives

615.747
**Sensors and
Sensor Systems**
FA / SP • Asynchronous

615.763
**Infrared Sensors:
Physics and Applications**
SP • Asynchronous



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